

Schur Pseudo-Likelihood

Scoring and Regularizing Correlation in High Dimensions

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Abstract

We introduce the *Schur pseudo-likelihood*, a one-parameter family that damps the cross-block coupling of the Gaussian likelihood through its Schur complements, for *scoring* and regularizing covariance and correlation estimates in high dimensions. The Gaussian log-likelihood is the standard criterion, but when the dimension rivals the sample size it is governed by the smallest, least-identifiable eigenvalues of the estimate, and as a criterion for ranking or selecting estimates it becomes unreliable; damping the coupling restores it. The optimal damping has a closed form—the reliability of the coupling, a James–Stein shrinkage. On real crypto returns, choosing a shrinkage estimate by the Schur pseudo-likelihood rather than by the full likelihood yields several-fold lower out-of-sample portfolio variance once the dimension rivals the sample size.

1 The high-dimensional failure of the Gaussian likelihood

A correlation matrix is the hard part of a covariance matrix: variances are a one-dimensional, per-series problem, while the high-dimensional difficulty—the near-singular spectrum and the unstable inverse—lives in the correlation. The standard way to judge a correlation (or covariance) estimate $\hat{R}$ is the log-likelihood it assigns to held-out data, $\ell(\hat{R}) = -\frac{1}{2}(\log \det \hat{R} + \text{tr}(\hat{R}^{-1}S))$ for a test covariance S . Both terms are governed by the smallest eigenvalues of $\hat{R}$: $\log \det \hat{R} = \sum_i \log \lambda_i$ diverges as any $\lambda_i \rightarrow 0$, and $\hat{R}^{-1}$ diverges with it. When p rivals n those eigenvalues are unidentifiable (the Marčenko–Pastur regime), so the likelihood is dominated by the least reliable part of the spectrum [Ledoit and Wolf, 2004, Bun et al., 2017]. As a *criterion*, it then fails: in a controlled experiment in which the truth is known, the held-out likelihood ranks the better of two estimates below chance, whereas inversion-free criteria such as Frobenius distance and the variogram score [Scheuerer and Hamill, 2015] exceed 80%. This is the evaluation-side analogue of the instability of the minimum-variance portfolio $w \propto \hat{R}^{-1}\mathbf{1}$, which fails through the same inverse. The remedy is to discount that inverse—continuously, through one interpretable parameter, rather than by discarding the cross-block structure outright.

2 The Schur pseudo-likelihood

Partition the variables into K ordered blocks. The Gaussian density factors exactly into block conditionals, $p(x) = \prod_k p(x_k | x_{<k})$, each Gaussian with covariance the Schur complement

$$S_k = R_{kk} - R_{k,<k} R_{<k,<k}^{-1} R_{<k,k}, \quad \mu_{k|<k} = R_{k,<k} R_{<k,<k}^{-1} x_{<k},$$

so $\log \det R = \sum_k \log \det S_k$ and the likelihood is a sum of block-conditional terms, each inverting only the conditioning block [Vecchia, 1988, Katzfuss and Guinness, 2021]. We damp the coupling by $\gamma \in [0, 1]$,

$$S_k(\gamma) = (1 - \gamma) R_{kk} + \gamma S_k, \quad \mu_{k|<k}(\gamma) = \gamma R_{k,<k} R_{<k,<k}^{-1} x_{<k},$$

and score $\ell_\gamma(R; x) = \sum_k \log \mathcal{N}(x_k; \mu_{k|<k}(\gamma), S_k(\gamma))$. At $\gamma = 1$ this is the exact joint likelihood; at $\gamma = 0$ it is the block-diagonal composite likelihood [Besag, 1975, Lindsay, 1988, Varin et al., 2011]; intermediate γ inverts no block larger than a conditioning set and is better conditioned. The same γ interpolates an allocation rule, from hierarchical risk parity at $\gamma = 0$ [López de Prado, 2016] to minimum variance at $\gamma = 1$ [Cotton, 2024].

3 The optimal damping is the coupling reliability

In a tractable two-block model— $x_2 = bx_1 + \varepsilon$ with conditional R^2 equal to ρ^2 —minimising the expected predictive residual of the damped score gives a closed form,

$$\gamma^\star = \frac{b^2}{b^2 + \text{Var}(\hat{b})} = \frac{(n-2)\rho^2}{(n-2)\rho^2 + (1-\rho^2)},$$

the *reliability* of the estimated coupling: a Wiener/James–Stein shrinkage [James and Stein, 1961, Ledoit and Wolf, 2012]. It tends to 1 as $n \rightarrow \infty$ or $\rho^2 \rightarrow 1$ and to 0 as $\rho^2 \rightarrow 0$, and is interior when the coupling is partially reliable. The empirical optimum tracks it: sweeping a synthetic block correlation’s coupling strength v against the aspect ratio n/p (Figure 1), the γ minimising recovery error rises monotonically with v , from 0 (no coupling) to 1 (strong coupling), and dips where samples are scarcest—the reliability law, recovered across the plane.

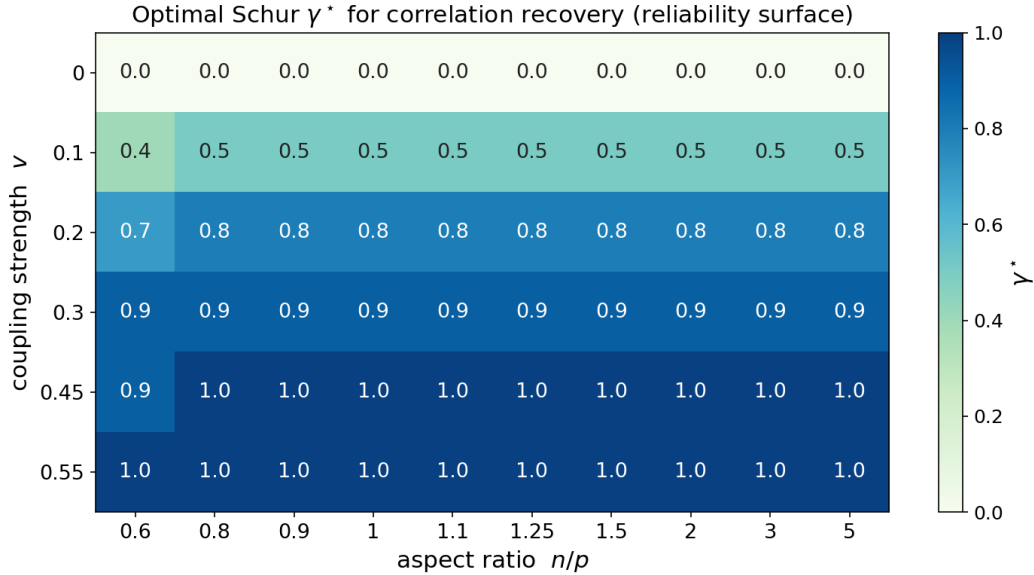


Figure 1: Optimal Schur $\gamma^\star$ over coupling strength v and aspect ratio n/p ($p = 128$, synthetic). It rises monotonically with v —the reliability law—and dips where samples are scarcest.

4 Selecting estimates in high dimensions: real data

The decisive test of a *scoring rule* is whether the estimate it prefers generalizes. We test it on real daily crypto returns (72 liquid assets) with a same-frequency, disjoint-split design that avoids both look-ahead and the cross-frequency attenuation (Epps) that confounds naive hourly-versus-daily comparisons: within a window, days are partitioned at random into a training set (an estimate is built), a validation set (scoring rules

choose among a shrinkage ladder $R_\delta = (1 - \delta)R_{\text{emp}} + \delta I$, and a held-out test set. The external criterion is the realized variance of the minimum-variance portfolio formed from the selected estimate—neither scoring rule, so the comparison is not circular.

We compare every scoring rule that is usable out of sample: the full likelihood ($\gamma = 1$), the Schur pseudo-likelihood, the block/composite likelihood ($\gamma = 0$), Stein loss, the variogram score, and—as a matched, non-circular benchmark, since validation and test are disjoint—the validation GMV variance itself. A clear split emerges (Figure 2). When $n/p \leq 1$ the *inverse-heavy* scores select too little shrinkage and build fragile portfolios: the full likelihood and Stein loss give median out-of-sample variance roughly six times higher than the rest. The *inverse-discounting* scores—the Schur pseudo-likelihood, the composite likelihood, and the matched GMV criterion—select near-optimally; Schur matches the composite likelihood (window-clustered bootstrap difference within ± 0.01) and comes within a few percent of the matched benchmark, while beating the full likelihood, Stein, and variogram with 95% CIs excluding parity. All converge once $n/p > 1.5$, where the full likelihood becomes reliable. That Schur *ties* the composite endpoint here is the reliability law at work: crypto returns are dense and factor-driven, so for arbitrary blocks the coupling is barely reliable, γ^* is small, and the dial lands near $\gamma = 0$. The interior- γ advantage of Schur over the composite endpoint appears when the coupling is partially reliable—the synthetic regime of the previous section. The robust message is that discounting the inverse is what matters in high dimensions, and the Schur dial is the principled, closed-form way to do it, spanning composite and full likelihood with $\gamma^* = \text{reliability}$.

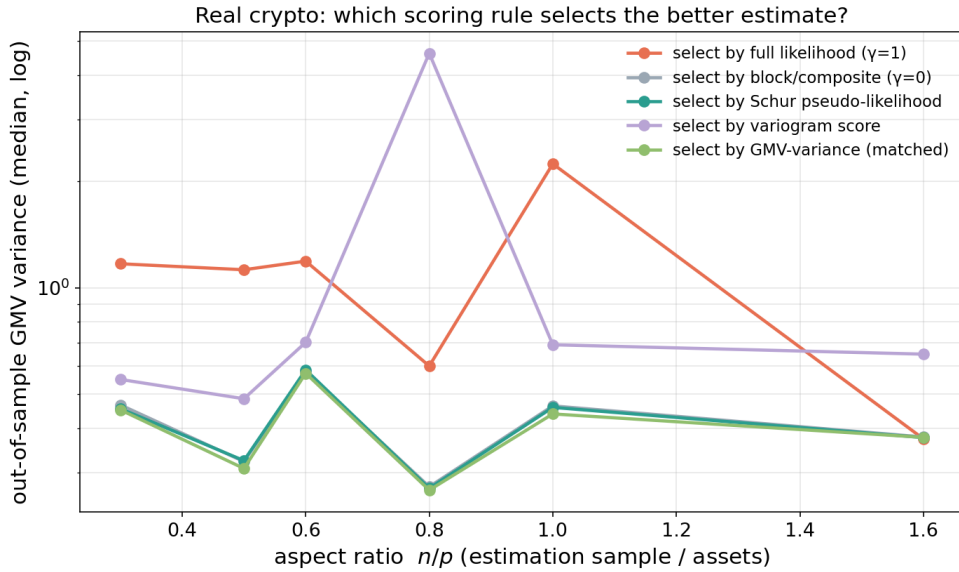


Figure 2: Real crypto. Out-of-sample minimum-variance-portfolio variance (median, log scale) of the estimate *selected* by each scoring rule, versus aspect ratio. Selecting by the Schur pseudo-likelihood is several-fold safer than by the full likelihood when undersampled; they converge once well sampled.

5 Regularization and conditioning

The damping is a structured shrinkage: $S_k(\gamma) = (1 - \gamma)R_{kk} + \gamma S_k$, a convex combination of positive-definite matrices, pulls each conditional covariance toward its unconditional block. This raises the smallest eigenvalues and lowers the condition number relative to the full likelihood, and yields a usable inverse even when the empirical correlation is singular ($n < p$); the block-diagonal endpoint is positive definite whenever the

blocks are. Where the linear interpolation can itself approach singularity—near a vanishing Schur complement under strong coupling—the affine-invariant SPD-geodesic interpolation lifts the small eigenvalues multiplicatively and stays well conditioned by construction. This conditioning is the method’s general high-dimensional benefit: the same γ that maximises the pseudo-likelihood regularizes any downstream use of the inverse, the evaluation-side analogue of eigenvalue shrinkage [Ledoit and Wolf, 2004, 2012, Bun et al., 2017].

6 Scope: a scoring and regularization device, not a free estimator

The maximiser of ℓ_γ recovers the true predictive law for every γ while implying an inflated correlation, so ℓ_γ is a scoring and regularization device, not an estimation objective. Used *as* an estimator the same damping helps recovery only in its own regime—a genuinely block-structured correlation with partially-reliable coupling (synthetic), where interior γ beats both endpoints—and not on dense, strongly factor-driven data such as crypto, where the empirical correlation recovers best and arbitrary block damping discards real signal. The optimal damping also inverts with the target: recovering the matrix wants large γ , recovering its inverse (partial correlations) wants small γ , because the inverse amplifies coupling noise. “Best” is undefined until the target is named; the contribution here is a principled, closed-form dial that is right for scoring and regularizing the inverse-fragile, high-dimensional case.

7 Conclusion

The Schur pseudo-likelihood is a one-knob interpolation between decoupled and full Gaussian scoring whose optimal setting is the closed-form reliability of the cross-block coupling. In high dimensions, where the Gaussian likelihood is dominated by unidentifiable small eigenvalues and fails as a criterion, it is a better-behaved, better-conditioned score: selecting estimates by it yields markedly safer out-of-sample portfolios on real data when the dimension rivals the sample size.

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